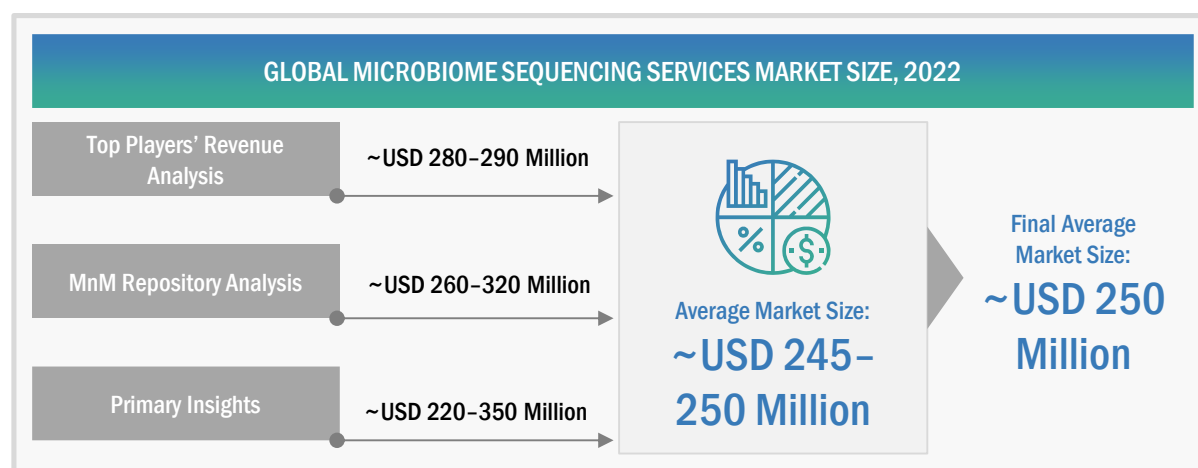


2.2 MARKET ESTIMATION METHODOLOGY

The report presents a detailed assessment of the microbiome sequencing services market and qualitative inputs and insights from MarketsandMarkets. The global size of the microbiome sequencing services market was estimated through multiple approaches. A detailed market estimation approach was followed to estimate and validate the value of the microbiome sequencing services market and other dependent submarkets, as mentioned below:

FIGURE 3 MICROBIOME SEQUENCING SERVICES MARKET SIZE ESTIMATION
(SUPPLY-SIDE ANALYSIS), 2022

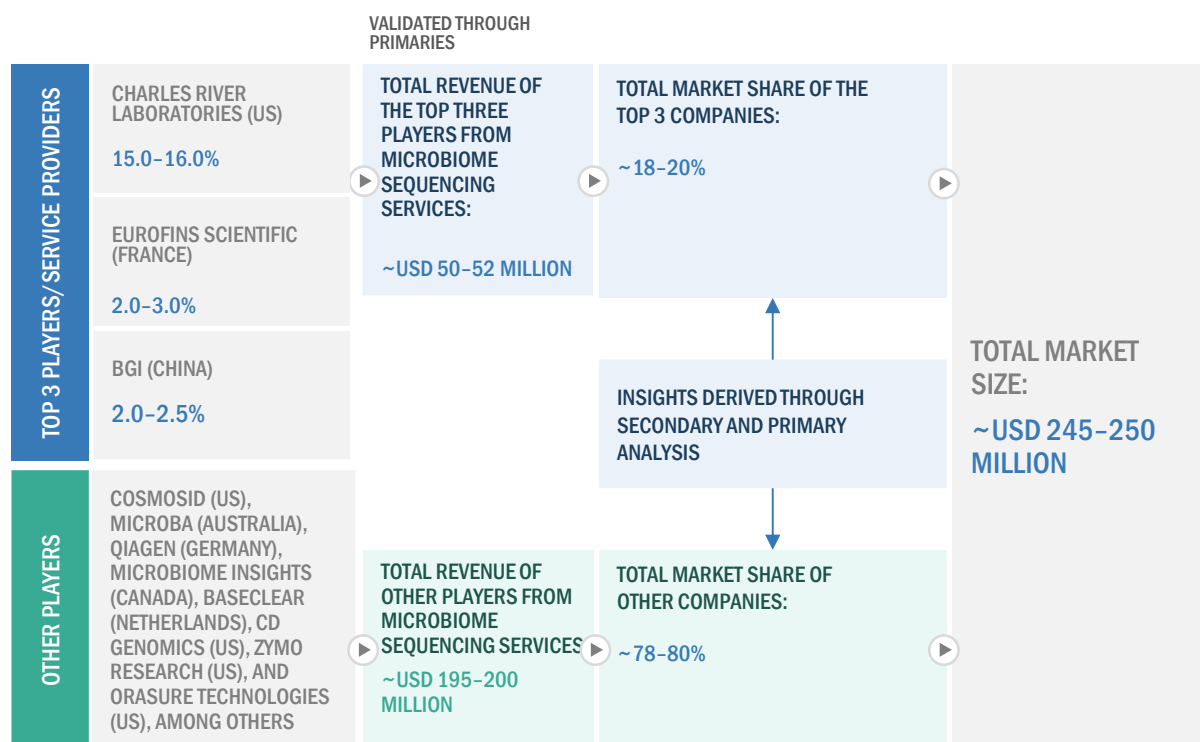


Source: MarketsandMarkets Analysis and Primary Insights

APPROACH 1: COMPANY REVENUE ANALYSIS (BOTTOM-UP APPROACH)

- Revenues for individual companies were gathered from public sources and databases.
- The microbiome sequencing services-related business shares of leading players were gathered from secondary sources to the extent available. In certain cases, the shares of microbiome sequencing services businesses were ascertained through a detailed analysis of various parameters, including service portfolios, market positioning, and primary insights.
- Individual shares or revenue estimates were validated through expert interviews.

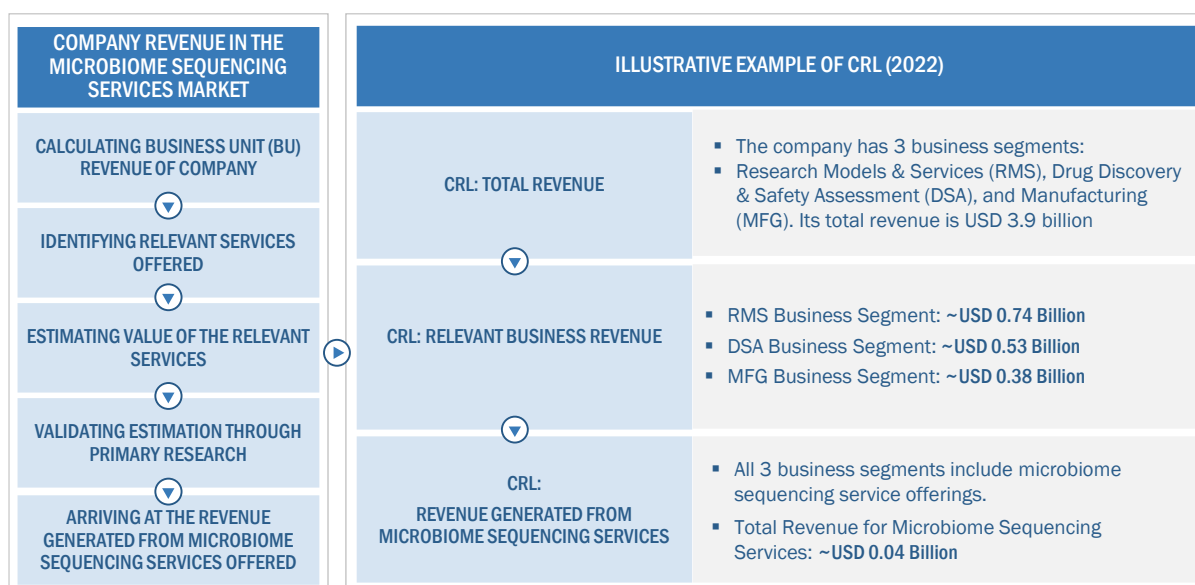
FIGURE 4 MARKET SIZE ESTIMATION: APPROACH 1 (COMPANY REVENUE ANALYSIS-BASED ESTIMATION), 2022



Source: MarketsandMarkets Analysis and Primary Insights

For the calculation of the global market value, the segmental revenue was arrived at based on the revenue mapping of major players active in the microbiome sequencing services market. This process involved the following steps:

- A list of the major global players operating in the microbiome sequencing services market was generated.
- The annual revenues generated by major global players from the microbiome sequencing services market, or the nearest reported business unit/category, were mapped.
- The revenues of the top 3 players in the microbiome sequencing services market were identified.
- The revenue of the top 3 players was assumed to be approximately 18-20% in 2022 and added up to 100% to derive the global value of the microbiome sequencing services market.

FIGURE 5 ILLUSTRATIVE EXAMPLE OF COMPANY REVENUE ANALYSIS, 2022

Source: Annual Reports, SEC Filings, Investor Presentations, Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

APPROACH 2: MNM REPOSITORY ANALYSIS

- For estimating the microbiome sequencing services market, MnM reports, such as the contract research organization (CRO) services market, next-generation sequencing (NGS) market, and metagenomic sequencing market, were considered. The global and regional market values of microbiome sequencing services and the dependent submarkets were extracted from the MnM repository and validated through secondary and primary research. The final global market size was triangulated through the average of all approaches and validated through primary interviews with industry experts.

APPROACH 3: SECONDARY RESEARCH

- The size of the microbiome sequencing services market was obtained from secondary sources.
- The shares of various microbiome sequencing services in the overall market were obtained from secondary data and validated through primary participants to arrive at the total microbiome sequencing services market.
- Primary participants further validated the numbers.

APPROACH 4: PRIMARY INTERVIEWS

- As part of the primary research process, individual respondents were interviewed for insights on market size and market growth.
- All responses were collated, and a weighted average was taken to derive a probabilistic estimate of the market size and growth rate.

2.2.1 INSIGHTS FROM PRIMARY EXPERTS

FIGURE 6 MARKET SIZE VALIDATION FROM PRIMARY SOURCES

FACTORS AFFECTING MARKET GROWTH									
 Product Manager	 The current market is growing bigger in size as new players are entering the market with varied services for microbiome sequencing.								
 Group Leader	 The key sequencing methods popular for microbiome profiling include amplicon sequencing and shotgun sequencing.								
 Research Scientist	 Emerging technologies such as nanopore sequencing are set to provide lucrative opportunities for market growth.								
MARKET TRENDS	ACCORDING TO A MAJORITY OF INDUSTRY EXPERTS, THE TOP MARKET PLAYERS ACCOUNTED FOR ~20% OF THE MICROBIOME SEQUENCING SERVICES MARKET								
R&D investments in oncology applications are positively impacting the market. Growing adoption of new sequencing tools is expected to drive service adoption. <i>– Market Experts</i>	<table> <tr> <th>COMPANY</th><th>MARKET RANK (2022)</th></tr> <tr> <td>CRL (US)</td><td>1</td></tr> <tr> <td>Eurofins Scientific (France)</td><td>2</td></tr> <tr> <td>BGI (China)</td><td>3</td></tr> </table>	COMPANY	MARKET RANK (2022)	CRL (US)	1	Eurofins Scientific (France)	2	BGI (China)	3
COMPANY	MARKET RANK (2022)								
CRL (US)	1								
Eurofins Scientific (France)	2								
BGI (China)	3								

Source: Primary Insights

The final market size was estimated based on the weighted average of values arrived at through various approaches, such as company revenue analysis, MnM repository analysis, secondary research, and primary insights. The research methodology and market numbers were further validated from primaries.

SEGMENT ASSESSMENT (BY SERVICE, TYPE, TECHNOLOGY, AND END USER)

The segmental assessment was conducted using the following approaches:

- Approach for service: Revenue contributions/splits of microbiome sequencing services were calculated from the total revenues of leading players in the market (wherever available), and respective growth trends were derived by studying the adoption patterns of various services considered in the study.
- Approach for type: Revenue contributions/splits of different segments (amplicon sequencing, shotgun sequencing, and whole-genome sequencing) to the total revenues of leading players in the market (wherever available) and respective growth trends were derived by studying the adoption patterns of various products considered in the study.
- Approach for technology: Splits for segments were derived by studying the adoption preference (based on the type of service, sample size, and turnaround time) for different technologies used in services, including sequencing by synthesis (SBS), sequencing by ligation (SBL), and nanopore sequencing. This information was validated through primary interviews.
- Approach for end users: Splits for segments were derived by studying the adoption patterns of microbiome sequencing services by different end users in the market.

- Approach for regions: The geographic assessment was conducted by studying geographic revenue contributions/splits of leading players in the market (wherever available) and respective growth trends.

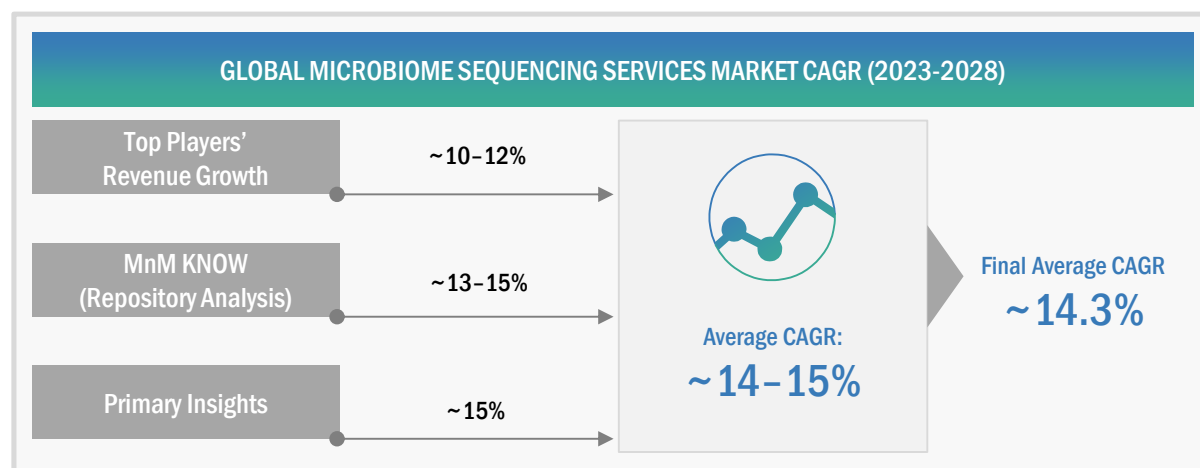
The assumptions and approaches were validated at each point through industry experts contacted during primary research. Considering the limitations of data available from secondary research, revenue estimates for individual companies (for the overall microbiome sequencing services market and geographic market assessment) were ascertained based on a detailed analysis of their respective service offerings, geographic reach/strength (direct or through distributors or suppliers), and shares of the leading players in a particular region or country.

2.3 MARKET GROWTH RATE PROJECTIONS

The growth forecast model was defined based on an assessment of the qualitative and quantitative factors affecting the market growth. These mainly included:

1. Historic revenue growth trends of the leading players in the market
 - Demand-side factor assessment, including clinical trials conducted for microbiome therapeutics, pharmaceutical R&D spending, the incidence of various diseases, the number of drug discovery and development projects across all geographies, and trial density, among other factors
 - Impact assessment of the key factors, such as drivers, restraints, challenges, and opportunities, influencing market growth
 - Impact assessment of the historic competitive trends driving market growth—service launches, strategic growth initiatives, innovation trends, and R&D initiatives
 - Impact assessment of regulatory guidelines/mandates on market growth over the forecast period. Only the defined regulatory guidelines related to the study period were considered for this assessment. The impact of any future regulatory uncertainties on market growth is not accounted for.
 - The impact of the recession on the overall market was evaluated for 2023–2028

FIGURE 7 MICROBIOME SEQUENCING SERVICES MARKET (SUPPLY-SIDE): CAGR PROJECTIONS



Source: MarketsandMarkets Analysis and Primary Insights